MILITARY SPECIFICATION

AIRCRAFT WEAPONS RELEASE SET

AN/AWE-1

This specification has been approved by the Naval Air Systems Command, Department of the Navy.

1. SCOPE

1.1 Scope - The equipment covered by this specification shall provide a programmed series of output release pulses to aircraft weapons stations. Release of a variety of stores from the aircraft in accordance with quantity, mode, and drop interval can be selected by the pilot.

1.2 Classification - Aircraft Weapons Release Set AN/AWE- I shall consist of the items listed in 6.9.

2. APPLICABLE DOCUMENTS

2.1 General - The documents listed in 6.11 of the issue in effect on the date of invitation for bids form a part of this specification to the extent specified herein.

3. REQUIREMENTS

3.1 Preproduction - This specification makes provision for preproduction testing.

3.2 Parts and Materials - In the selection of parts and materials, fulfillment of major design objectives shall be the prime consideration. In so doing the following shall govern:

(1) Microelectronic items shall conform to requirements specified herein.

(2) Other parts and materials requirements shall conform to Specification MIL-E-5400.

(3) Nonrepairable subassemblies, as outlined in Specification MIL-E-5400, shall be used when practicable. The general size of the subassembly and the amount of circuitry to be included therein shall be approved by the procuring activity. Nonrepairable subassemblies must be reliable (see 6. 4).

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(4) When previously produced models of this equipment did not use nonrepairable subassemblies, the design shall not be changed to employ nonrepairable assemblies without the approval of the procuring activity.

3.2.1 Nonstandard Parts and Materials Approval - Approval for the use of nonstandard parts and materials (including transistors and diodes) shall be obtained as outlined in Specification MIL-E-5400.

3.2.2 Microelectronic Modular Assemblies - when used, microelectronic modular assemblies shall be described by a contractor procurement specification and must be approved by the procuring agent y. Microelectronic modular assemblies (microelectronic circuits) (MICS) shall be contained within maintenance modules (for example, on printed circuit cards). The choice of microelectronic modular assemblies shall be based on at least two sources of supply, which can be used interchangeably.

3.2.3 Modules, Maintenance - The electronic portions of the equipment shall be divided into maintenance modules (see Specification MIL-S-23603). Maintenance modules shall normally be considered repairable.

3.2.4 Microelectronic Interchangeabilityy - Microelectronic circuits (MICS) shall be interchangeable at the MIC level, or at the maintenance module level when approved by the procuring activity (see 6. 13).

3.3 Design and Construction - The equipment shall conform with all the applicable requirements of Specification MIL-E-5400 for design, construction and workmanship, except as otherwise specified herein.

3.3.1 Total Weight - The total weight of the equipment, excluding cables, shall be a minimum consistent with good design and shall not exceed 9.75 pounds.

3.302 Reliability -

3.3.2.1 Operational Stability - The equipment shall operate with satisfactory performance, continuously or intermittently for a period of at least 1,000 hours without the necessity for readjustment of any controls which are inaccessible to the operator during normal use.

3.3.2.2 Operating Life - The equipment shall have total operating life of 10,000 hours with reasonable servicing and replacement of parts. Parts requiring scheduled replacement shall be specified by the contractor.

3.3.2.3 Reliability in Mean Time Between Failure (MTBF) - The equipment shall have 2,000 hours of mean (operating) time between failures when tested and accepted as outlined under the requirements of 4.4.3.

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3.3.3 Cabling and Connections -

3.3.3.1 Cables and Connectors - The equipment shall provide for the use of cables and connectors in accordance with Specification MIL-E-5400.

3.3.3.2 Interconnection Cabling - The equipment shall be capable of satisfactory operation using external wiring in accordance with the applicable requirements of Specification MIL-W-5088 and EI-458. The external wiring shall be unshielded, except that a minimum number of the individual wires may be shielded when demonstrated as necessary to meet interference control requirements and provided the assembly of the cable to its plugs may be easily accomplished. External cables and that portion of the connectors attached to the cables shall not be supplied as part of the equipment. (See figure 1.)

3.3.4 Control Panels - All rack or console mounted control panels shall conform to the applicable requirements of Specification MIL-C-6781. The configuration of all control panels must be approved by the procuring activity prior to preproduction testing.

3.3.5 Interchangeability of Reordered Equipment - For reordered equipment interchangeability y shall exist between units and all replaceable assemblies, subassemblies, and parts of a designated model of any previously manufactured equipment, supplied or designated by the procuring activity.

3.3.6 Interference Control - The generation of radio interference by the equipment and the vulnerability y of the equipment to radio interference shall be cent rolled within the limits of Specification MIL-I-6181.

3.3.7 Maintenance Provisions and Field Testing - Provisions for maintenance shall be as specified in Specification MIL-E-5400. Specific test points and test facilities shall be provided to the greatest extent practicable for ease of field testing and maintenance.

3.3.8 Nomenclature and Nameplates - Nomenclature assignment and nameplate approval for equipment identification shall be in accordance with Specification MIL-N-18307.

3.3.9 Standard Conditions - The following conditions shall be used to establish normal performance characteristics under standard conditions and for making laboratory bench tests.

Temperature	Room ambient (25 ±5°C)
Altitude	Normal ground
Vibration	None
Humidity	Room ambient up to 90% relative humidity
Input power voltage	115 ±1.0 vac and/or 27.5 ±0.5V DC

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3.3.10 Service Conditions - The equipment shall operate satisfactorily under of the environmental service conditions or reasonable combination of these conditions as specified in Specification MIL-E-5400 for Class II equipment, except as modified herein:

3.3.10.1 Vibration - The equipment shall operate satisfactorily when subjected to the vibration requirements of Curve I of Specification MIL-E-5400.

3.3.10.2 Altitude - The equipment shall operate satisfactorily when subjected to the altitude requirements of Specification MIL-E-5400 with the exception that the operational altitude be limited to 50,000 feet.

3.3.11 Warm-Up Time - The equipment shall require no warm-up time.

3.3.12 Primary Input Power Requirements - The equipment shall meet all applicable requirements of MIL-STD-704 and shall give specified performance from the following power sources with characteristics as defined in MIL-STD-704 having limits as specified therein. The power required shall not exceed the specified amounts.

(1) AC Power (Single Phase) 115V, 400Hz, Category “B”, 10 VA

(2) AC Power (Single Phase) 5V, 400Hz, Category “B”, 5 VA

(3) DC Power 28V, Category “B”, 0.25 amp, 7 Watts

3.3.12.1 Degraded Performance - Degraded performance will be permitted for voltage transients not exceeding O. 5 second during normal electric system operation. Operation shall return to normal with no resulting damage to the equipment.

3.4 Performance - Unless otherwise specified, values set forth to establish the requirements for satisfactory performance apply to performance under both standard and extreme service conditions. When reduced performance under the extreme conditions is acceptable, tolerances or values setting forth acceptable variations from the performance under the standard conditions will be specified.

3.4.1 Operation - The system, upon depression of the weapons release button, shall provide a selectable quantity of output release signals to any number of weapon stations, up to a maximum of eight wing stations and one center station, that may exist aboard the aircraft. The stations that are to receive the release signals are those selected by the pilot on the aircraft's station select panel, not supplied as part of this equipment.

3.4.2 Modes of Release - The system herein described shall generate release signals, as described in 3.4.1, in the followlng manner:

(1) To cause stores drop in ripple or step pairs (one from port and one from starboard).

(2) To cause stores drop in ripple or step singles (one from port and then one from starboard).

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(3) To cause stores drop in ripple or step salvo (one from each selected station simultaneously).

When the system is in the ripple mode the described drops will continue at a selected drop interval described herein until the pilot has released the desired amount of stores (see 3.4.4). When the system is in the step mode the described drops will occur at a rate determinedly the pilot. This rate will be one release pulse generated for each pilot depression of the weapon release button.

3.4.2.1 Ripple Pairs -When the ripple pairs mode is selected and one or more pairs of stations of equal inboard to outboard priority are selected the system will ripple stores simultaneously from each of the two highest priority stations selected (brie port, one starboard) up to the quantity selected. If the amount selected is greater than the capacity of the pair, the station transfer signals will cause the system to ripple stores from the next highest priority pair selected. If there is no next highest priority pair selected, the system will not generate any further fire pulses.

3.4,2.2 Ripple Singles - The ripple singles mode shall release one store from the highest priority station selected during one output pulse: the next output pulse shall release one store from the opposite wing station of equal inboard to outboard priority if selected. If the opposite, equal priority station was not selected, the fire pulse shall be continuously directed to the highest priority station until it is completeIy depleted of stores if required by the quantity selection. If after depletion, more stores are required to be dropped, the station transfer signals shall cause drops to occur from the next highest priority station selected. At this priority level, the drop shall again alternate between the same priority opposite wing stations if selected. If not, the drops shall continue to be directed to the single station as previously described in this paragraph.

3.4.2.3 Ripple Salvo - The ripple salvo mode shall cause the simultaneous release of one store from each station selected until the quantity of stores selected has been released. There is no regard to priority in the salvo mode.

3.4.2.4 Step Pairs - For the step pairs mode, there shall be a simultaneous release from both highest priority stations (similar inboard to outboard priority) for each depression of the weapon release button. Priority and station transfer conditions shall remain the same as those described in 3.4.2.1.

3.4.2.5 Step Singles - For the step singles mode, there shall be a single release from the highest priority station selected for the first depression of the weapon release button. The next depression shall cause a release from the equal priority station on the opposite wing, if selected. These conditions of priority and station transfer are the same . as those described in 3.4.2.2.

3.4.2.6 Step Salvo -The step salvo mode shall cause simultaneous release of one store from each station selected, provided it contains at least one store, for each depression of the weapon release button.

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3.4.3 Priority of Release - The system herein described shall generate release originals, as described in 3.4.1 in a priority manner as follows:

(1) Outboard stations shall have priority over inboard stations. As long as an outboard station is selected and contains stores, it shall receive the release signals before any more inboard selected stations, regardless of its port of starboard location.

(2) Opposite wing stations of equal outboard to inboard priority shall receive release signals alternately in step singles until one of the stations is depleted. At this time the release pulses will be directed to the other station and continue until that station is depleted. Then priority reverts inboard in the same manner.

For maximum configuration the station priority shall be as follows:

Station (Port and Starboard)	Priority
1 and 8	1
2 and 7	2
3 and 6	3
4 and 5	4
Center	5

3.4.4 Quantity Select - The number of release output pulses generated by the system shall be pilot controlled. The pilot when desiring a drop of any amount of stores, shall be able to select the exact amount of from 2 to 40 output pulses to be generated in amounts limited to the following:

(1) 1, 2, 3, 4, 5, 6 (steps of one)

(2) 8, 12, 16, 20 (steps of four)

(3) 30, 40 (steps of ten)

When in the singles mode, one store will be depleted for each output pulse. In the pairs mode, two stores will be depleted for each output pulse. In the salvo mode, the number of stores depleted for each output pulse will be equal to the number of loaded stores stations selected.

3.4.5 Rate of Release - The interval between successive release output pulses in the ripple mode shall be pilot controlled and continuously variable between 20 and 2,000 milliseconds. The output pulse interval shall be within ±5 percent of the control setting for standard conditions and ±10 percent for extreme conditions. In the step mode, each release pulse is initiated by a depression of the weapon release button.

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3.4.6 System By-Pass - The system shall provide a by-pass mode which will enable the pilot to release stores when the programmer is in the off position. This will provide a step release from selected stations, for each depression of the weapon release button. An optional external jumper cable can be supplied with each system to enable normal stores release if it becomes necessary to remove the programmer from the aircraft for any reason.

3.4.7 Pulse Width - The duty cycle of the output pulses shall be 50 percent of the output pulse interval described in 3.4.5 with accuracies as follows:

Pulse Interval	Accuracy
20-50 ms	±40 pct
50-2000 ms	±10 pct

In the step mode, the pulse width shall be determined by the pilot. The leading edge of the release pulse shall occur, when the pilot manually starts a step drop. The trailing edge of the" pulse shall occur when the pilot manually ends the drop.

3.4.8 System Tone Closure - The system shall provide a closure for an external audio tone when stores are released. In ripple, the closure shall be provided at the start of a quantity drop and be removed when the last store of the quantity has been released (see 3.4.4). In the step mode the closure shall be provided at the same time and during the same duration as the pulse width described in 3.4.7.

3.4.9 Firing Current - During the firing pulse, the system shall be able to provide 10 amperes at 28 volts dc to each station being fired. The firing pulse will be of a duration as described in 3.4.7.

3.4.10 Leakage Current - The current, except during actual firing, shall be zero.

3.4.11 System Outputs - The system shall provide fire pukes, at the proper current as specified in 3.4.9.

3.4.12 Fail-Safe Provisions - Fail-safe provisions shall be incorporated into the design of the system such that any failure in the output driver stages shall not result in any type of spurious release signals.

3.4.13 Single Test Point Provision - A single test connector shall be provided in the system such that all out put driver circuits can be interrogated from this one point when the system is installed in the aircraft.

3.4.14 Aircraft Weapons Inventory Provisions - A single test connector shall be provided in the system such that all output driver circuits can be monitored for weapons release.

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3.5 Detail Requirement

3.5.1 Control, Aircraft Weapons Release Set C-6682(*)/AWE-1 - The control unit shall meet the requirements of Specification MIL-C-81360(AS) and be capable of controlling the following functional factors: (See figure *2.)*

3.5.1.1 Function - The control is the unit of the system which selects the operating modes of the system. The control shall determine the following:

3.5.1.1.1 Quantity Select - The number of stores to be released.

3.5.1.1.2 Drop Interval - The time period between the release of stores.

3.5.1.1.3 Multiplier - The increasing of the drop interval by a factor of ten or unity.

3.5.1.1.4 Mode Outputs - The patterns or methods in which stores are released. The release of stores is as follows, with the modes of release as described in 3.4.2.

3.5.1.1.4.1 Step Singles - The release of a single stores as desired by a manually actuated release button.

3.5.1.1.4.2 Step Pairs - The release of a pair of stores by a manually actuated release button.

3.5.1.1.4.3 Step Salvo - The simultaneous release of stores from all stations by a manually actuated release button.

3.5.1.1.4.4 Ripple Singles - The release of a selected quantity of single stores at a selected drop interval.

3.5.1.1.4.5 Ripple Pairs - The release of a selected quantity of pairs of stores at a selected drop interval.

3.5.1.1.4.6 Ripple Salvo - The simultaneous release of a selected quantity of of stores from all stations at a selected drop interval.

3.5.2 Programmer, Aircraft Weapons Release Set MX-7016(*)/AWE-1 - The programmer shall meet the requirements *of* Specification MIL-P-81361(AS) and be capable of controlling the followlng functions: (See figure 2.)

3.5.2.1 Function - The programmer is the unit of the system which generates the fire pulse train required to release stores from the aircraft in accordance with commands received from the weapon release stitch, control, and stores information that may be available from the aircraft weapons stations. The unit shall be capable of operating from one to nine stores racks by the proper addition of loglc modules to the basic package composed of *the* following:

3.5.2.1.1 Logic Power Supply - This power supply provides all logic level voltages required by the programmer.

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3.5.2.1.2 Station Transfer Logic - Transfer logic provides release of stores from the highest priority stations first, then automatically changes to the next lower priority station(s) selected and continues releasing stores until the selected number of stores have been released.

3.5.2.1.3 Output Drivers - A driver is provided for each weapon station to supply the necessary firing pulse(s) for stores release.

3.5.2.1.4 Quantity Select Logic - This logic limits the number of ripple output pulses to the quantity selected by the pilot at the control.

3.5.2.1.5 Fire Pulse Counter - The programmer provides a fire pulse counter for each weapon station. If a rack is providing an identification signal indicating a multiple store (2 through 6 rounds) and this signal is not removed by the time 6 fire pulses are counted, the fire pulse counter itself will generate the station transfer signal. If a rack is providing an indication signal indicating a multiple store (7 or more rounds) the fire pulse counter circuit will be disabled.

3.5.3 Indicator, Aircraft Weapons Release Set ID-1310(*)/AWE-1 - The indicator shall meet the following requirements:

3.5.3.1 Function - The indicator shall provide an instantaneous store indication, in decimal form, of the quantity of stores remaining at any weapon station at any time.

3.5.3.2 Form Factor - The indicator shall conform to the dimensions shown in Figure 4. It shall be mounted in the cockpit.

3.5.3.3 Weight - The weight of the indicator shall not exceed 2 pounds.

3.5.3.4 Contents - The indicator shall contain a decimal counting unit (0-9) for each station. The count indication shall drop by one digit each time the unit is pulsed by the output of the programmer.

3.5.3.5 Controls - Each decimal unit of the stores inventory display is to be manually preset to any number available (0-9) directly at the panel itself.

3.5.3.6 Electrical Connections - A single connector shall be provided for electrical connection to the programmer.

3.6 Design Data - No data is required by this specification (other than reports accompanying samples submitted for testing) or by applicable documents, unless specified in the contract or ordered (see 6.7).

4, QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government, The Government

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reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Classification of Tests - Items covered by this specification shall be subjected to the following tests to determine compliance with all applicable requirements:

(1) Preproduction Tests

(2) Initial Production Tests

(3) Acceptance Tests

(4) Life Tests

4.2 Preproduction Tests - Preproduction tests shall be made on an equipment representative of the production equipments to be supplied under the contract. Preproduction tests shall be accomplished under the responsibility of the contractor and shall be conducted in accordance with the approved test procedure of 4.6. The government inspector and the procuring activity shall be advised when tests are to be conducted so that a representative may be designated to witness or supervise the tests when so desired. Contractors not having adequate facilities to conduct all required tests shall obtain the services of a commercial testing laboratory acceptable to the government.

4.2.1 Preproduction Test Data - The contractor shall submit all data collected in conducting these tests to the procuring agency for review and approval. On first production contracts this data shall include a list of all electrical and electronic parts giving their specified voltage, current and temperature rating and the applied circuit voltage, current and ambient and surface temperatures. The ambient and surface temperature shall be obtained under the extreme high temperature operating condition.

4.2.2 Scope of Tests - Preproduction tests shall include all tests deemed necessary by the procuring activity to determine that the equipment meets all the requirements of this specification and the contract. Preproduction tests shall include environmental tests in accordance with the procedures of Specification MIL-T-5422 and interference tests in accordance with Specification MIL-I-6181.

4.2.3 Preproduction Approval - Approval of the preproduction sample shall be by the procuring activity upon satisfactory completion of all tests. No production equipments shall be delivered prior to the approval of the preproduction sample. Prefabrication of production equipment prior to the approval of the preproduction sample is at the contractor's own risk. The approved preproduction sample shall be retained by the contractor for his use in the fabrication and testing of equipment to be submitted for acceptance. The preproduction sample shall not be considered as one of the equipments under the contract.

4.2.4 Production Equipments - Equipments supplied under the contract shall in all respects, including design, construction, workmanship, performance and quality, be equivalent to the approved preproduction sample. Each equipment shall be capable of successfully passing the same tests as imposed on the preproduction sample. Evidence of

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non-compliance with the above shall constitute cause for rejection and for equipment already accepted by the Government, it shall be the obligation of the contractor to make necessary corrections as approved by the procuring activity.

4.3 Initial Production Tests - One of the first ten production equipments shall be selected and sent at the contracto's expense to a designated Government laboratory for tests. This equipment shall be selected by the procuring activity after the equipment has successfully passed all individual tests. The preproduction sample shall not be selected for this test.

4.3.1 Scope of Tests - This equipment may be subjected to any and all tests the procuring activity deems necessary to assure that the production equipment is equivalent to the previously approved preproduction sample in design, construction, workmanship, performance and quality and that it meets all applicable requirements.

4.3.2 Accessory Material - In addition to the complete equipment submitted for initial production tests the contractor shall also submit such accessory material and data necessary to test the equipment.

4.3.3 Initial Production Sample Approval - Approval of the initial production sample shall be by the procuring activity upon satisfactory completion of all tests. Any design, material or performance defect made evident during this test shall be corrected by the contractor to the satisfaction of the procuring activity. Failure of the initial production sample to pass any of the tests shall be cause for deliveries of equipment under the contract to cease until proper corrective action is approved and accomplished. Corrective action shall also be accomplished on equipment previously accepted when requested by the procuring activity.

4.3.4 Reconditioning of Initial Production Test Sample - On completion of the initial production test the equipment shall be reworked by the contractor by replacing all wear or damaged items. After reworking, the contractor shall resubmit the equipment for acceptance.

4.4 Acceptance Tests - The contractor shall furnish all samples and shall be responsible for accomplishing the acceptance tests. All inspection and testing shall be under the supervision of the government inspector. Contractors not having testing facilities satisfactory to the procuring activity shall engage the service of a commercial testing laboratory acceptable to the procuring activity. The contractor shall furnish test reports showing quantitative results for all acceptance tests. Such reports shall be signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Acceptance tests shall consist of the following:

(1) Individual Tests

(2) Sampling Tests

(3) Reliability Assurance Tests

(4) Special Tests

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4.4.1 Individual Tests - Each equipment submitted for acceptance shall be subjected to the individual tests. These tests shall be adequate to determine compliance with the requirements of material, workmanship, operational adequacy and reliability. As a minimum, each equipment accepted shall have passed the following tests:

(1) Examination of Product

(2) Operational Test

(3) Manufacturing Run In Test

4.4.1.1 Examination of Product - Each equipment shall be examined carefully to determine that the material and workmanship requirements have been met.

4.4.1.2 Operational Test - Each equipment shall be operated long enough to permit the equipment temperature to stabilize and to check sufficient characteristics and record adequate data to assure satisfactory equipment operation.

4.4.1.3 Manufacturing Run In Test - Each equipment shall be operated under the conditions specified herein for a period of 6 hours without failure. A failure shall be defined as anything which causes malfunctioning of the equipment. Only those adjustments will be permitted which can be made by using such controls and adjustments that are accessible to the operator during the normal use of the equipment.

Temperature	Ambient room
Humidity	Ambient room
Vibration	Any selected frequency within the range of 20 to 30 Hz (excluding resonant points) and a minimum amplitude of ±3 g's

The equipment shall be vibrated (without vibration isolators) for a period of 10 minutes prior to the beginning of the 6-hour period of operation. Where feasible, the equipment shall be operated during this vibration period for the purpose of detecting flaws and imperfect workmanship. operation within the specified limits of satisfactory performance is not necessarily required during the vibration period. The direction of vibration should be vertical to the normal mounting plane for 5 minutes and lateral to the plane for 5 minutes, Where it is not feasible to vibrate the equipment in 2 directions, the vertical direction shall be used. During the 6-hour period of operation followlng the 10-minute vibration period, the equipment shall be mechanically cycled periodically through its various phases of operation. Should a failure occur, it should be repaired and the test started over, except that the 10-minute vibration period need not be repeated when it is certain the failure was not a result of the vibration. Should repetitive failures occur, corrective action shall be taken to eliminate this defect from future equipment. A record shall be kept of all failures. The 6-hour period epecified above may be composed of two 3-hour periods to conform with standard working hours.

4.4.2 Sampling Tests - Equipments selected for sampling tests shall first have passed the individual tests. Equipments shall be selected for sampling tests by the government inspector in accordance with the following: (Sampling tests shall not be conducted unless reliability assurance tests of 4.4.3 are deleted by contractual action.)

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Quantity of Equipments Offered for Acceptance	Quantity to be Selected For Sampling Test
First 10	0*
Next 50	1
Next 75	1
Next 100	1
	1 for each additional 200 or fraction thereof

*NOTE: If by contract action the initial production test (4, 3) is deleted, then a sample test shall be conducted on one equipment from the first 10 produced.

4.4.2.1 Scope of Tests - As a minimum, each equipment selected for sampling tests shall be subjected to the following tests:

(1) Complete operational test at ambient room conditions, making all necessary measurements to assure that all applicable specification requirements have been met.

(2) Operational test at certain environmental conditions. The conditions may vary for each equipment tested and should be based on results of the preproduction, initial production, individual and special tests.

(3) Manufacturing run in test specified in 4.4.1.3 except that the test duration shall be 120 hours with no restriction on the number of failures. However, each failure shall be analyzed as to cause and remedial action necessary to reduce the possibility of its recurrence in future equipment.

4.4.3 Reliability Assurance Tests - Reliability assurance tests shall be conducted as required by MIL-STD-781. Equipments selected for reliability assurance tests shall first have passed the individual tests.

4.4.3.1 Test Level - The Test Level shall be F from Table I of MIL-STD-781.

4.4.3.2 Accept-Reject Criteria - Test Plan V of MIL-STD-781 shall be used to determine the accept-reject criteria for the reliability y qualification test plan. Test Plan VI of MIL-STD-781 shall be used to determine the accept-reject criteria for the reliability sampling test plan.

4.4.3.3 Length of Heat Portion of Cycle - After stabilization at the high temperature limit required by the test level, the equipment shall be operated 2 hours during each cycle.

4.4.3.4 Performance Characteristics to be Measured - The performance characteristics to be measured shall be as specified in 6.10.

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4.4.3.5 Failure Criteria - In addition to the requirements of MIL-STD-781, the following requirements shall be used to determine when a failure has occurred during the test:

(1) Whenever performance characteristics fall below the acceptance requirement (4.4.3.4) at least one failure has occurred. If subsequent analysis reveals that several parts have deteriorated, each shall be counted a failure, unless one caused the other parts to fail.

4.4.3.6 Preventive Maintenance - During the period of the tests no preventive maintenance measures are required.

4.4.4 Special Tests - Special tests shall be conducted on a quantity of equipments for the purpose of checking the effect of any design or material change on the performance of the equipment and to assure adequate quality control. The equipment selected for special tests may be selected from equipments previously subjected to the sampling or reliability assurance tests.

4.4.4.1 Special Test Schedule - Selection of equipments for special tests shall be made as follows:

(1) On an early equipment after an engineering or material change.

(2) Whenever failure reports or other information indicate additional tests are required. (This will be determined by the procuring activity.)

4.4.4.2 Scope of Tests - Special tests shall consist of such tests as approved by the procuring activity. Test procedures previously approved for the preproduction tests shall be used where applicable. When not applicable, the contractor shall prepare a test procedure and submit It to the procuring activity for approval prior to conducting the tests.

4.4.5 Equipment Failure - Should a failure occur during either the sampling, reliability y assurance or special tests, the following action shall be taken:

(1) Determine the cause of failure.

(2) Determine if the failure is an isolated case or design defect.

(3) Submit to the procuring activity for approval, proposed corrective action intended to reduce the possibility of the same failure(s) occurring in future tests.

(4) Where practical, include a test in the individual test to check all equipment for this requirement until reasonable assurance is obtained that the defect has been satisfactorily corrected.

4.5 Life Test - The contractor shall furnish all samples and shall be responsible for accomplishing the life test. The test shall be of 300-hours duration and shall be conducted on equipments that have passed the individual test. The life test shall be performed

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under the conditions specified in 4.5.1. The life test sample shall be selected by the government inspector in accordance with the following. (Equipments which have successfully passed the initial production test, sampling tests, reliability tests, or special tests may be selected for life tests.) When reliability tests are conducted, the life test may be omitted if, during the reliability tests, a quantity of equipments equal to, or more than, that listed below receive at least 300 hours each of test time.

Quantity of Equipments Offered for Acceptance	Quantity to be Selected for Life Test
First 25	1
Next 175	1
Next 300	1
	1 for each additional 500 or fraction thereof

4.5.1 Test Conditions - The life test shall be conducted under the following simulated service conditions:

Temperature	Normal room
Altitude	Normal ground (0 - 5000 ft)
Humidity	Room ambient
AC Voltage	115 ±5V (at lowest applicable frequency)
DC Voltage	27.0 ±2.0V

4.5.2 Test Periods - The test may be run continuously or intermittently. Any period of operation shall be of sufficient duration to permit the equipment temperature to stabilize. Periodically, the equipment shall be turned on and off several times and put through its various phases of operation.

4.5.3 Performance Check - At approximately 8-hour intervals during the test, a limited performance check shall be made. The performance check proposed by the contractor shall be subject to approval by the procuring activity.

4.5.4 Test Data - The contractor shall keep a daily record of the performance of the equipment, making particular note of any deficiencies or failures. In the event of part failures, the defective parts shall be replaced and the operation resumed for the balance of the test period. A record shall be kept of all failures throughout the test, including all tube failures. This record shall indicate the following:

(1) Part type number

(2) The circuit reference symbol number

(3) The part function

(4) Name of the manufacturer

(5) Nature of the failure

(6) The number of hours which the part operated prior to failure

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4.5.4.1 Failure Report - In the event of a failure, the government Inspector shall be notified immediately. A report shall be submitted to the procuring activity upon completion of the test. In this report, the contractor shall propose suitable and adequate design or material corrections for all failures which occurred. The procuring activity will review such proposals and determine whether they are acceptable.

4.5.5 Reconditioning of Life Test Samples - Equipment which has been subjected to the life test shall be reconditioned as follows:

(1) On completion of the life tests, the equipment shall be reworked by the contractor by replacing all "wear" items. The "wear" items shall be determined by agreement between the contractor and the procuring activity.

(2) After reworking, the contractor shall resubmit the equipment for acceptance.

4.6 Test Procedures - The procedures used for conducting preproduction tests, acceptance tests and life tests shall be prepared by the contractor and submitted to the procuring activity for review and approval. The right is reserved by the procuring activity or the government inspector to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification or the contract. Specification MIL-T-18303 shall be used as a guide for preparation of test procedures. When approved test procedures are available from previous contracts such procedures will be provided and may be used when their use is approved by the procuring activity. However, the right is reserved by the procuring activity to require modification of such procedures, including additional tests, when deemed necessary.

4.7 Reconditioning of Tested Equipment - Equipment which has been subjected to acceptance and life tests shall be reconditioned by the contractor by replacing all wear or damaged items. After reworkhg the contractor shall resubmit the equipment for acceptance.

4.8 Presubmission Testing - No item, part or complete equipment shall be submitted by the contractor until It has been previously tested and inspected by the contractor and found to comply, to the best of his knowledge and belief, with all applicable requirements.

4.9 Rejection and Retest - Equipment which has been rejected may be reworked or have parts replaced to correct the defects and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished the government inspector.

5. PREPARATION FOR DELIVERY

5.1 General - All major units and parts of the equipment shall be preserved, packaged, packed and marked for the level of shipment specified in the contract or order In accordance with Specification MIL-E-17555.

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6. NOTES

6.1 Intended Use - This equipment is to be used in aircraft carrying stores to be released. This equipment can accurately control the rate of release, quantity to be released and the priority and mode of release in step or ripple. An optional feature of the system provides an instantaneous inventory of the quantity of stores remaining at any station at any time. The equipment shall meet the performance requirement of this specification when installed in accordance with EI-458 and meets the test requirements of ET-458.

6.2 Teat VaIues - Normal and limiting values of performance data shall be determined at input voltages of 27.5 ±0.5 vdc and 115 ±1.0 vac as applicable. These data are to be used in testing the equipment at installation points for compliance with minimum acceptance standard of performance.

6.3 Performance Objectives - Minimum size and weight, simplicity of operation, ease of maintenance, and an improvement in the performance and reliability of the specific functions beyond the requirements of this specification are objectives which shall be considered in the production of this equipment. Where it appears a substantial reduction in size and weight or improvement in simplicity of design, performance, ease of maintenance or reliability will result from the use of materials, parts and processes other than those specified in Specification MIL-E-5400, it is desired their use be investigated. When investigation shows advantages can be realized, a request for approval shall be submitted to the procuring activity for consideration. Each request shall be accompanied by complete supporting information.

6.4 As a general rule, nonrepairable subassemblies should be encapsulated or hermetically-sealed. The number of connections internal to the subassembly should be held to a minimum, Detail parts tolerances and ratings should be so selected that the life of the subassembly is greater than that of a similar repairable one. With few exceptions (such as high voltage power supplies), the nonrepairable subassembly should evidence a mean-time-to-failure greater than 5000 hours, and for many applications this figure must be nearer 50,000 hours.

6.5 Precedence of Documents - When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(1) Contract - The contract shall have precedence over any specification.

(2) This Specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, shall be specifically approved in writing by the procuring activity.

(3) Referenced Specifications - Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

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6.6 The parentheses (*), when used in the type designation, will be deleted or replaced by either a number or letter furnished by the procuring activity upon application by the contractor for assignment of nomenclature in accordance with 3.3.8. The complete type number shall be used on nameplates, shipping records and instruction books, as applicable.

6.7 Ordering Data - Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(1) Title, number, and date of this specification.

(2) Selection of applicable levels of packaging and packing (see 5.1).

(3) Data requirements (see 3.6).

6.8 Asterisk - In specification revisions and superseding amendments an asterisk "*" preceding a paragraph number denotes paragraphs in which changes have been made from the previous issue. This has been done as a convenience only and the government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content as written, irrespective of the asterisk notations and relationship to the last previous Issue.

6.9 Items of Equipment - The equipment covered by this specification shall consist of the following items:

Item	Type Designation	Appl. Para.
Control, Aircraft Weapons Release Set	C-6682(*)/AWE-l	3.5.1
Programmer, Aircraft Weapons Release Set	MX-7016(*)/AWE-l	3.5.2
Indicator, Aircraft Weapons Release Set	ID-1310(*)/AWE-l	3.5.3

6.9.1 Associated Equipment - The equipment shall operate in all aircraft in conjunction with presently existing aircraft armament systems, which include station select, master arm and weapon release switches when installed in accordance with EI-458. (See figure 1.)

6.10 Performance Characteristics to be Measured - The equipment shall meet the following performance characteristics.

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Characteristic	Specification requirements	Time in cycle that measurement is to be made
Modes of Release	3.4.2	
Priority of Release	3.4.3	During time temperature is stabilized
Quantity Select	3.4.4	
Rate of Release	3.4.5	

6.11 Applicable Documents -

SPECIFICATIONS

Military

MIL-W-5088	Wiring; Aircraft, Installation of
MIL-E-5400	Electronic Equipment, Aircraft, General Specification for
MIL-T-5422	Testing, Environmental, Aircraft Electronic Equipment
MIL-I-6181	Interference Control Requirements, Aircraft Equipment
MIL-C-6781	Control Panel: Aircraft Equipment, Rack or Console Mounted
MIL-E-17555	Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of
MIL-T-18303	Test Procedures; Preproduction and Inspection, for Aircraft Electronic Equipment, Format of
MIL-N-18307	Nomenclature and Nameplates for Aeronautical Electronic and Associated Equipment
MIL-S-23063	System Readiness/Maintain-ability; Avionic Systems Design, General Specification for

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STANDARDS

Military

MIL-STD-704	Electric Power, Aircraft Characteristics and Utilization of
MIL-STD-781	Reliability Tests Exponential Distribution

PUBLICATIONS

Naval Air Systems Command

EI-458	Installation Instructions for Aircraft Weapons Release Set AN/AWE-l
ET-458	Bench, Preflight, and Flight Test Instructions For Installation of Aircraft Weapons Release Set AN/AWE-l
MIL-C-81360(AS)	Control, Aircraft Weapons Release Set C-6682(*)/AWE-l
MIL-P-81361(AS)	Programmer, Aircraft Weapons Release Set MX-7016(*)/AWE-l

6.11.1 Availability of Documents -

(1) When requested specifications, standards, drawings, and publications refer to both title and number. Copies of this specification and applicable specifications required by contractors in connection with specific procurement functions may be obtained upon application to the Commanding Officer, Naval Supply Depot, Code 105, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120.

6.12 Criteria for Multiple Source - The criteria of this paragraph shall be used for determining the extent to which the requirement for multiple independent source availability of microcircuits is satisfied. For interchangeability at the MIC level, multiple source availability exists if the devices from the different sources may be used interchangeably, device by device, in the final equipment without any need for mechanical or electrical changes. For interchangeability at the card level, multiple source availability exists if a card composed of MICS from one source is mechanically and electrically interchangeable with another card composed of MICS from another source. Numerous aspects need be considered to arrive at the independence of sources; these involve production independence, corporate (or firm) and financial independence, and geographical separation. The production

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independence of one source of another is determined by the extent to which the one source is independent of the other for manufacture: and by the lack of common original suppliers of raw or processed materiaIs. Design, process information, technology, and diffusion masks may be derived from a common source. However, in such a case, each device source shall have a demonstrated capability of originating these items. The corporate (or firm) and financial independence of one source of another is determined by the absence of common directors, common officers, or holders of large common financial interests. Geographical separation is simply the distance between the locations where principal manufacture takes place by two sources.

6.13 Microelectronic Interchangeability Demonstration - Interchangeability of microelectronic elements shall be demonstrated as follows:

(1) For the interchangeability at the MIC level, it shall be demonstrated for each type of functional MIC from one source, as used in the system, that it is mechanically and electrically interchangeable with that from another source (see 6.12).

(2) For the interchangeability y at the Maintenance module level, it shall be demonstrated that modules typical of those to be used in the system composed of MICS from one source are mechanically and electrically interchangeable with modules composed of MICS from another source.

(3) For the completed equipment, it shall be demonstrated that modules composed of MICS from one source are interchangeable with corresponding modules composed of MICS from another source. When interchange has been effected, no degradation shall take place in system and subsystem performance.

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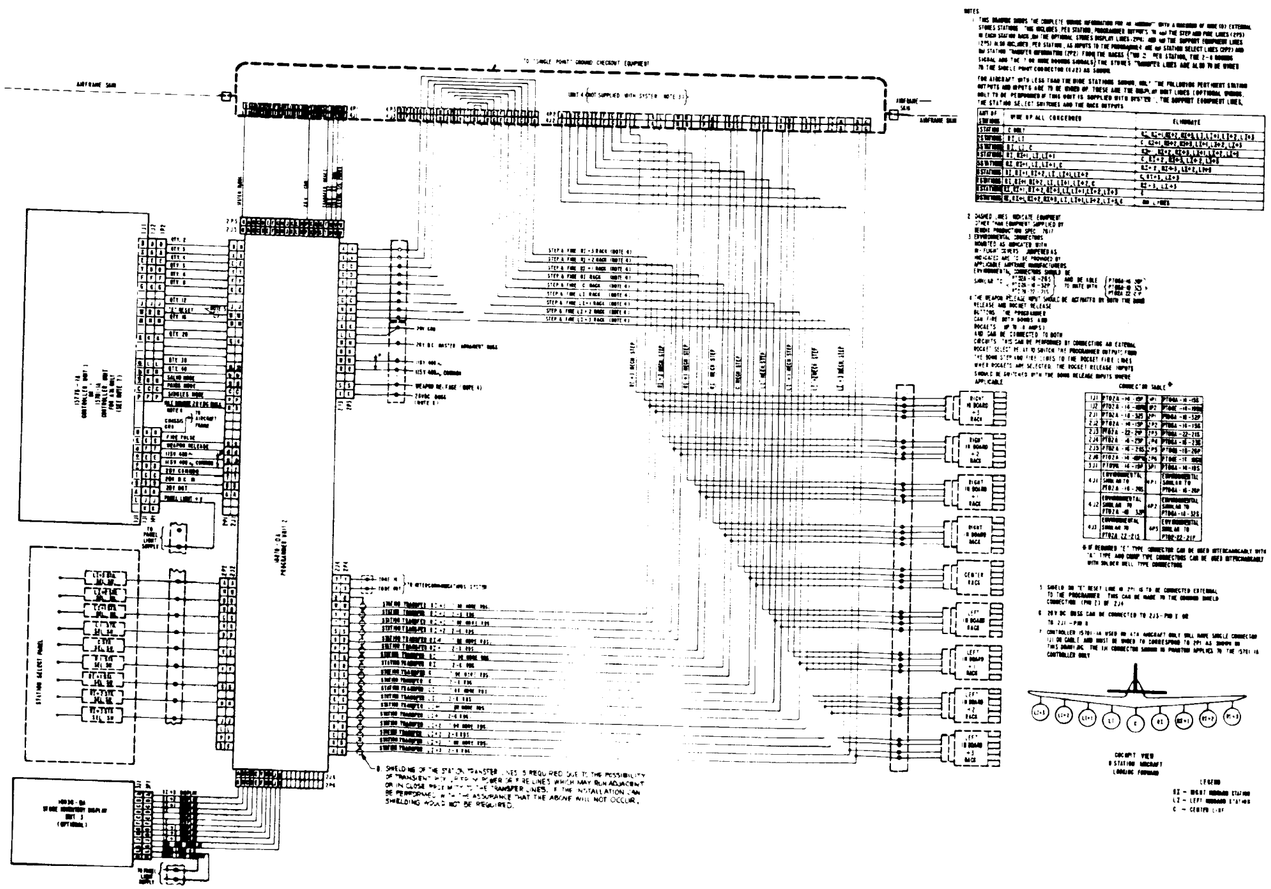


Figure 1. Interconnecting Wiring

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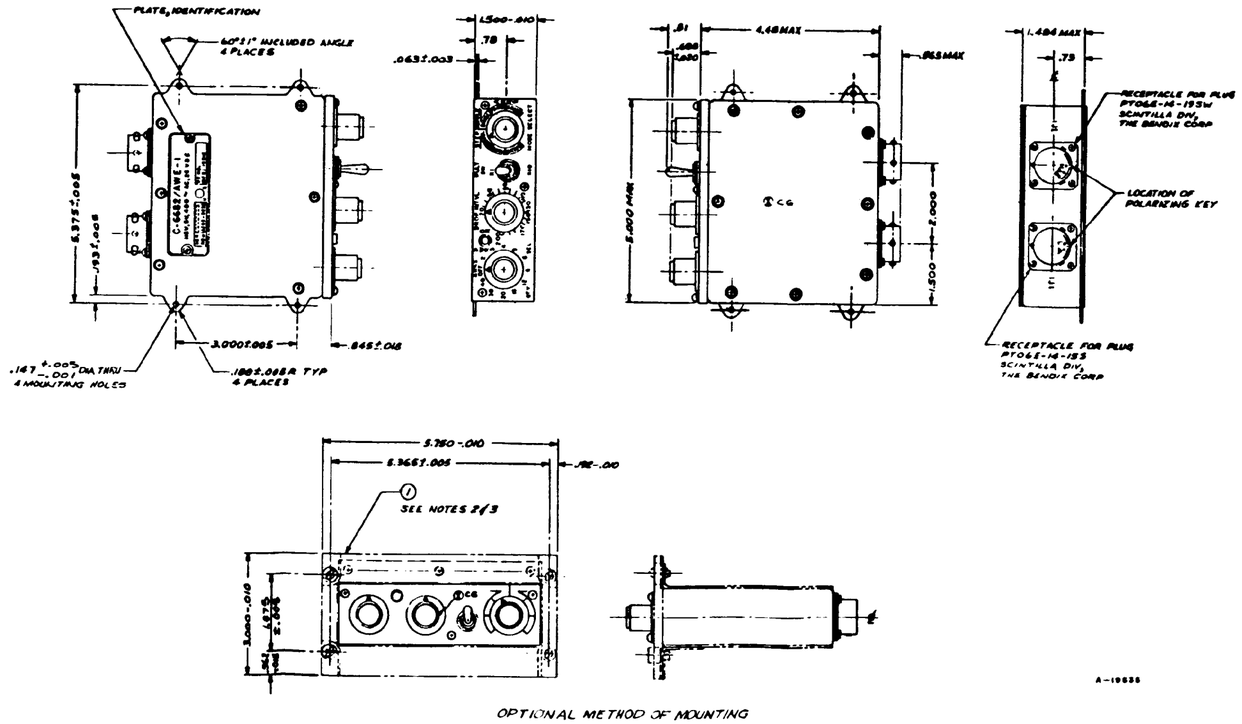


Figure 2. Outline Dimensions of Control

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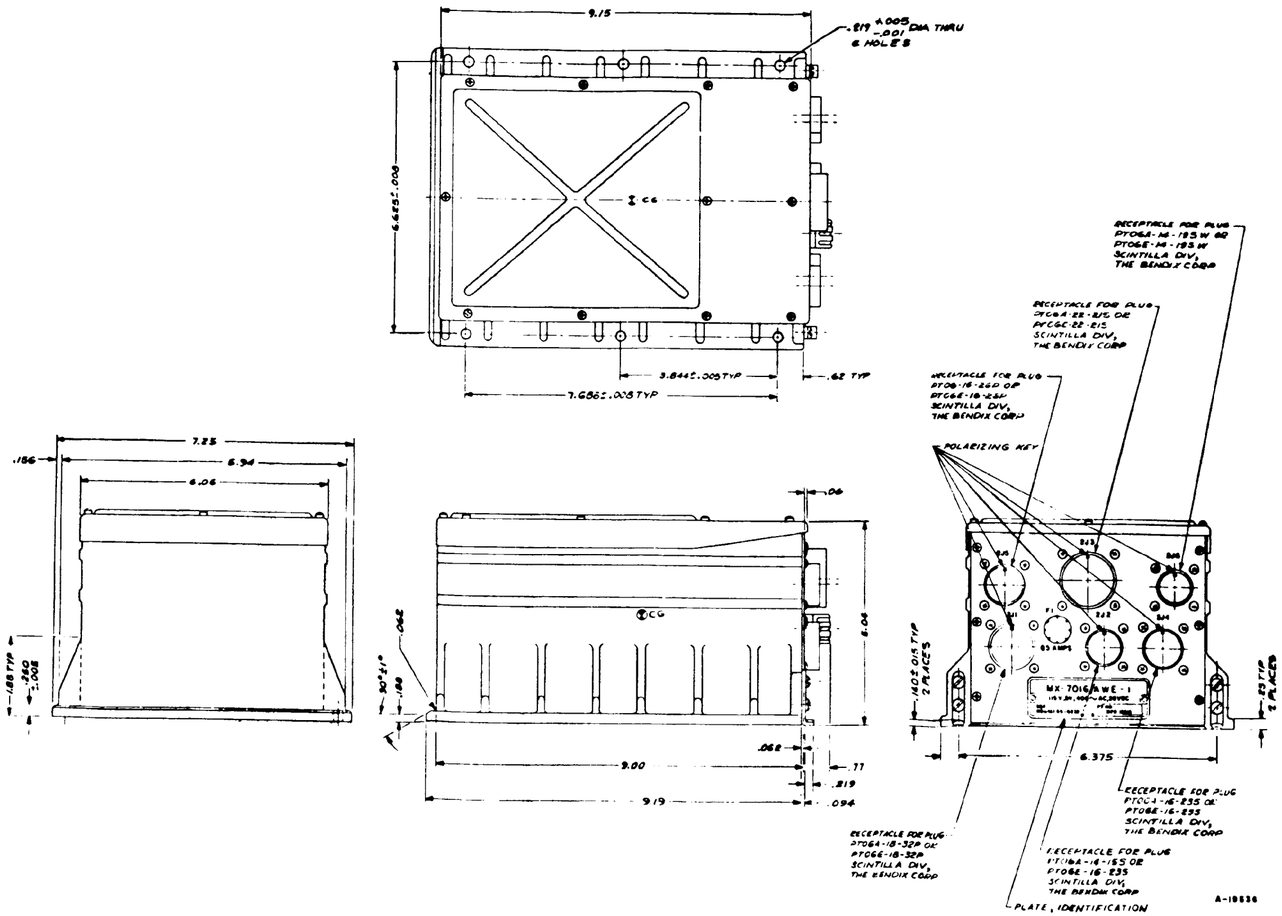
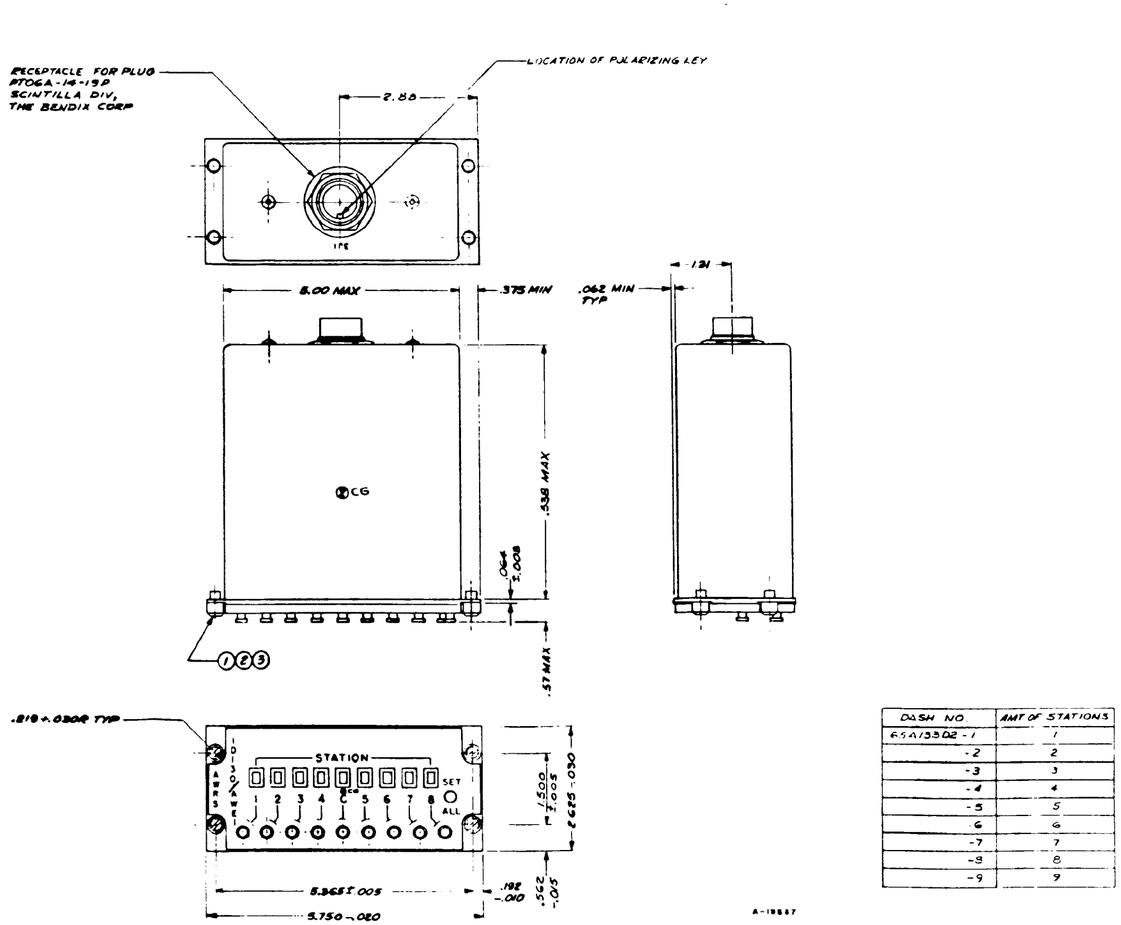


Figure 3. Outline Dimensions of Programmer

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DASH NO.	AMT OF STATIONS
65A13302-1	1
-2	2
-3	3
-4	4
-5	5
-6	6
-7	7
-8	8
-9	9

Figure 4. Outline Dimensions of Indicator

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION
MIL-A-81359(AS) AIRCRAFT WEAPONS RELEASE SETAN/AWE-1

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD, FORM 1426
APR 62

C-8870

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Naval Air Systems Command
Washington, D.C. 20360

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